

TE Curve Verification Pipeline Familywise ONNX Report - periodic_lstm_sequence

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory: `output/validation_checks/track2_familywise_onnx_report/periodic_lstm_sequence/2026-07-09-10-24-20__track2_periodic_lstm_sequence_familywise_onnx_report` ;
- summary YAML: `output/validation_checks/track2_familywise_onnx_report/periodic_lstm_sequence/2026-07-09-10-24-20__track2_periodic_lstm_sequence_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/periodic_lstm_sequence/2026-07-09-10-24-20__track2_periodic_lstm_sequence_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/periodic_lstm_sequence/2026-07-09-10-24-20__track2_periodic_lstm_sequence_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `periodic_lstm_sequence` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_periodic_lstm_sequence_fw_simplified_setpoints</code>	<code>2026-07-09-02-19-42__te_periodic_lstm_sequence_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_periodic_lstm_sequence_bw_simplified_setpoints</code>	<code>2026-07-09-02-32-24__te_periodic_lstm_sequence_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_periodic_lstm_sequence_global_simplified_setpoints</code>	<code>2026-07-09-02-10-37__te_periodic_lstm_sequence_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/periodic_lstm_sequence/forward/2026-07-09-02-19-42__te_periodic_lstm_sequence_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/periodic_lstm_sequence/forward/2026-07-09-02-19-42__te_periodic_lstm_sequence_fw_simplified_setpoints/python/periodic_lstm_sequence-epoch=095-val_mae=0.00348296.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/periodic_lstm_sequence/backward/2026-07-09-02-32-24__te_periodic_lstm_sequence_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/periodic_lstm_sequence/backward/2026-07-09-02-32-24__te_periodic_lstm_sequence_bw_simplified_setpoints/python/periodic_lstm_sequence-epoch=074-val_mae=0.00352408.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/periodic_lstm_sequence/global/2026-07-09-02-10-37__te_periodic_lstm_sequence_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/periodic_lstm_sequence/global/2026-07-09-02-10-37__te_periodic_lstm_sequence_global_simplified_setpoints/python/periodic_lstm_sequence-epoch=088-val_mae=0.00353329.ckpt</code>

Aggregate Metrics

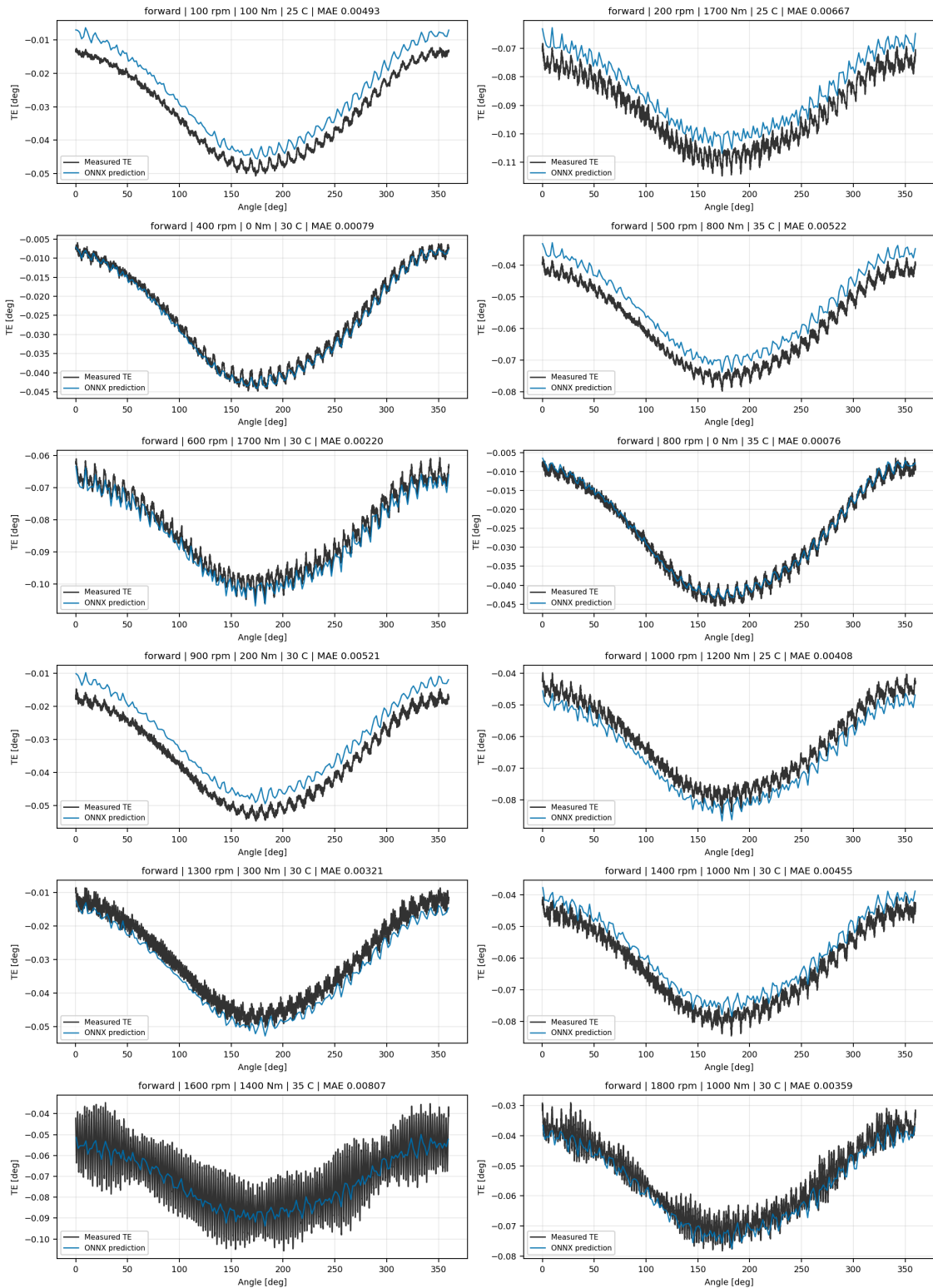
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003259	0.003506	7.645	13.799
backward	97	0.003346	0.003642	7.804	14.920
global	194	0.003369	0.003655	7.871	13.010

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000545	0.002918	0.001145	0.002279
backward	0.000464	0.002684	0.001467	0.003743
global	0.000032	0.002933	0.001339	0.003023

Forward 12-Curve Page

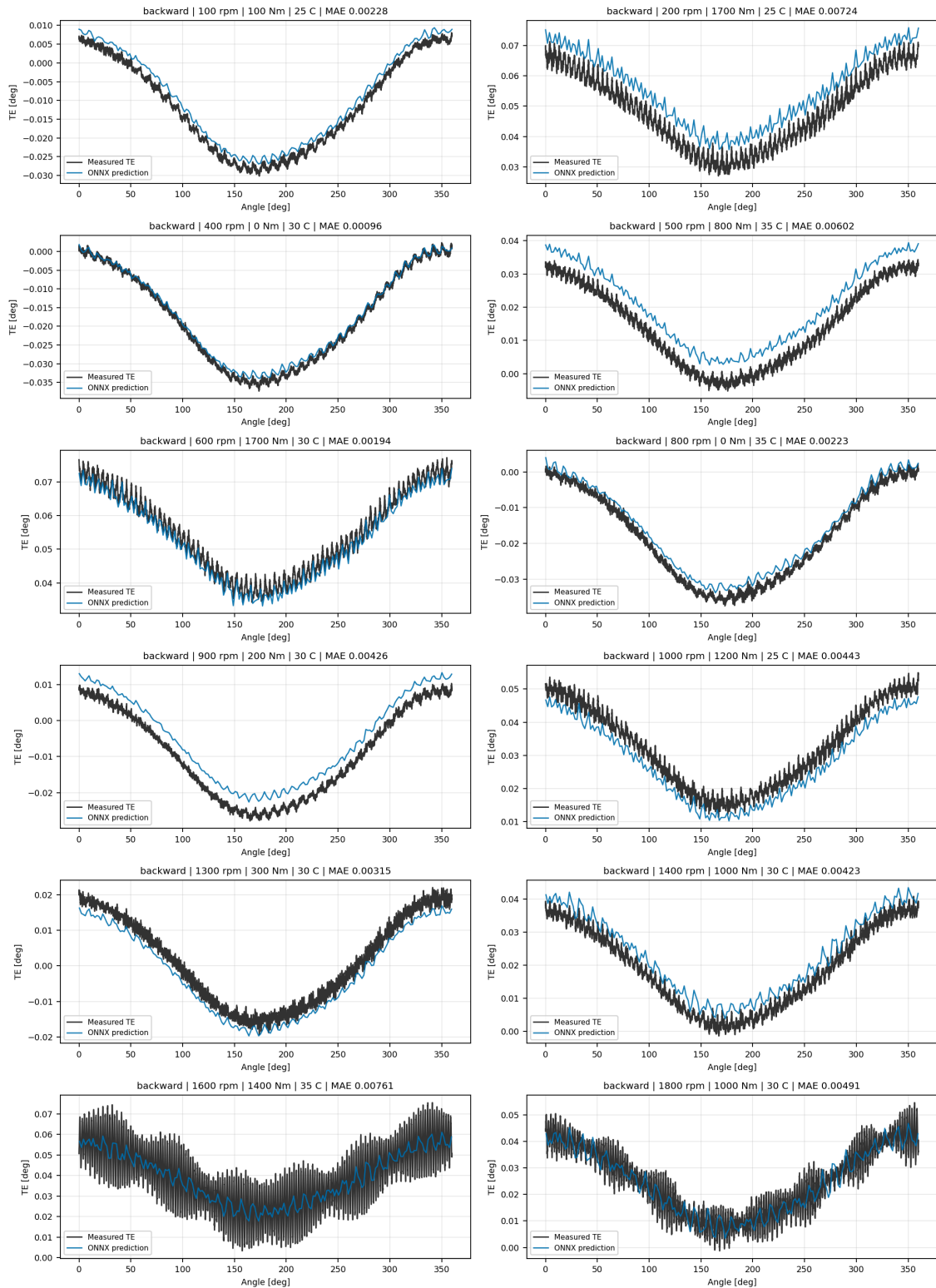
Simplified Dataset + Setpoints | periodic_lstm_sequence | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

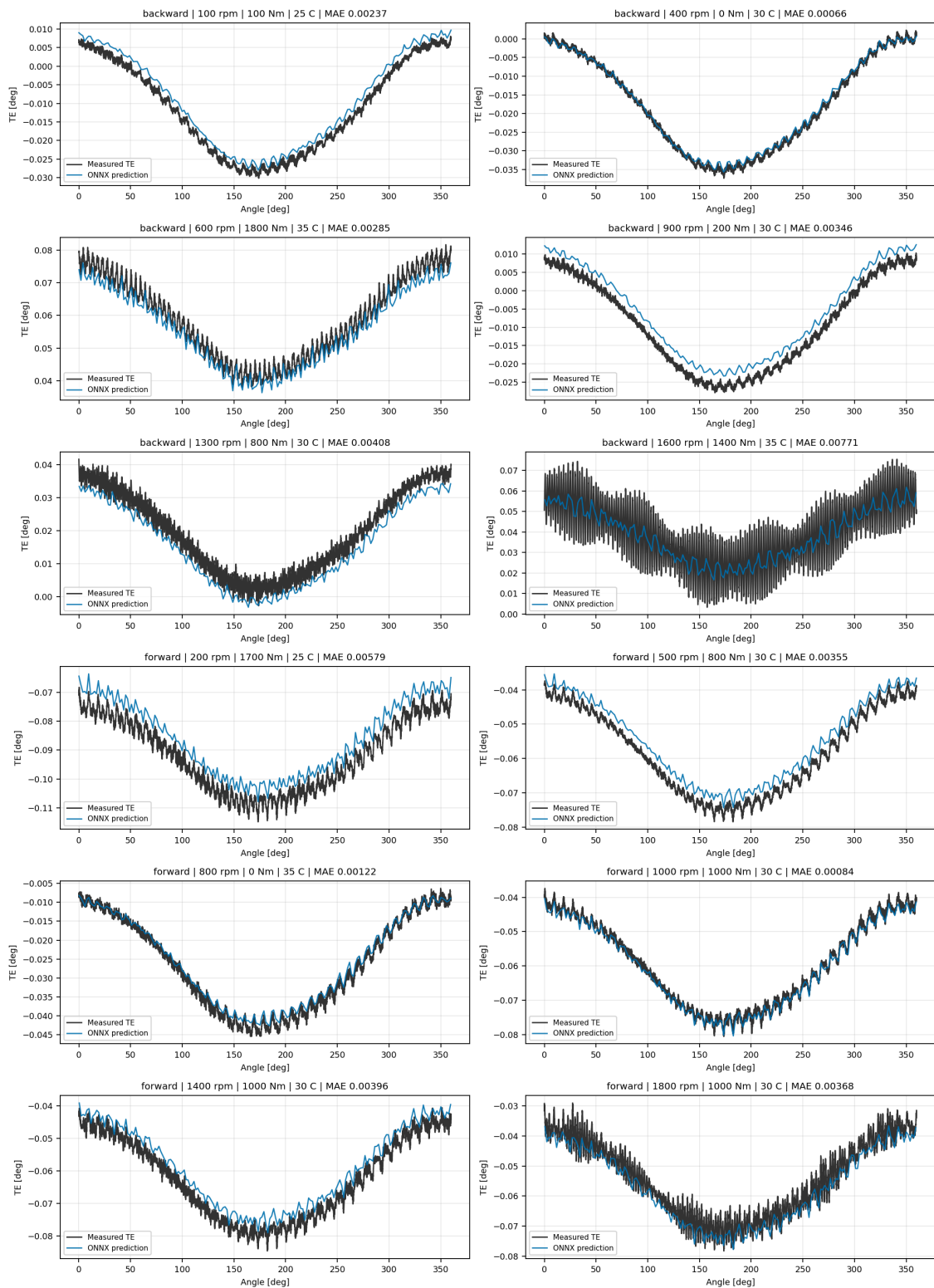
Simplified Dataset + Setpoints | periodic_lstm_sequence | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | periodic_lstm_sequence | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: periodic_lstm_sequence ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_periodic_lstm_sequence_fw_polished_setpoints	2026-07-09-03-42-29__te_periodic_lstm_sequence_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_periodic_lstm_sequence_bw_polished_setpoints	2026-07-09-03-56-20__te_periodic_lstm_sequence_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_periodic_lstm_sequence_global_polished_setpoints	2026-07-09-03-02-43__te_periodic_lstm_sequence_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/periodic_lstm_sequence/forward/2026-07-09-03-42-29__te_periodic_lstm_sequence_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/periodic_lstm_sequence/forward/2026-07-09-03-42-29__te_periodic_lstm_sequence_fw_polished_setpoints/python/periodic_lstm_sequence-epoch=047-val_mae=0.00186699.ckpt
backward	models/polished_dataset/setpoints/exported/periodic_lstm_sequence/backward/2026-07-09-03-56-20__te_periodic_lstm_sequence_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/periodic_lstm_sequence/backward/2026-07-09-03-56-20__te_periodic_lstm_sequence_bw_polished_setpoints/python/periodic_lstm_sequence-epoch=208-val_mae=0.00138862.ckpt
global	models/polished_dataset/setpoints/exported/periodic_lstm_sequence/global/2026-07-09-03-02-43__te_periodic_lstm_sequence_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/periodic_lstm_sequence/global/2026-07-09-03-02-43__te_periodic_lstm_sequence_global_polished_setpoints/python/periodic_lstm_sequence-epoch=245-val_mae=0.00137071.ckpt

Aggregate Metrics

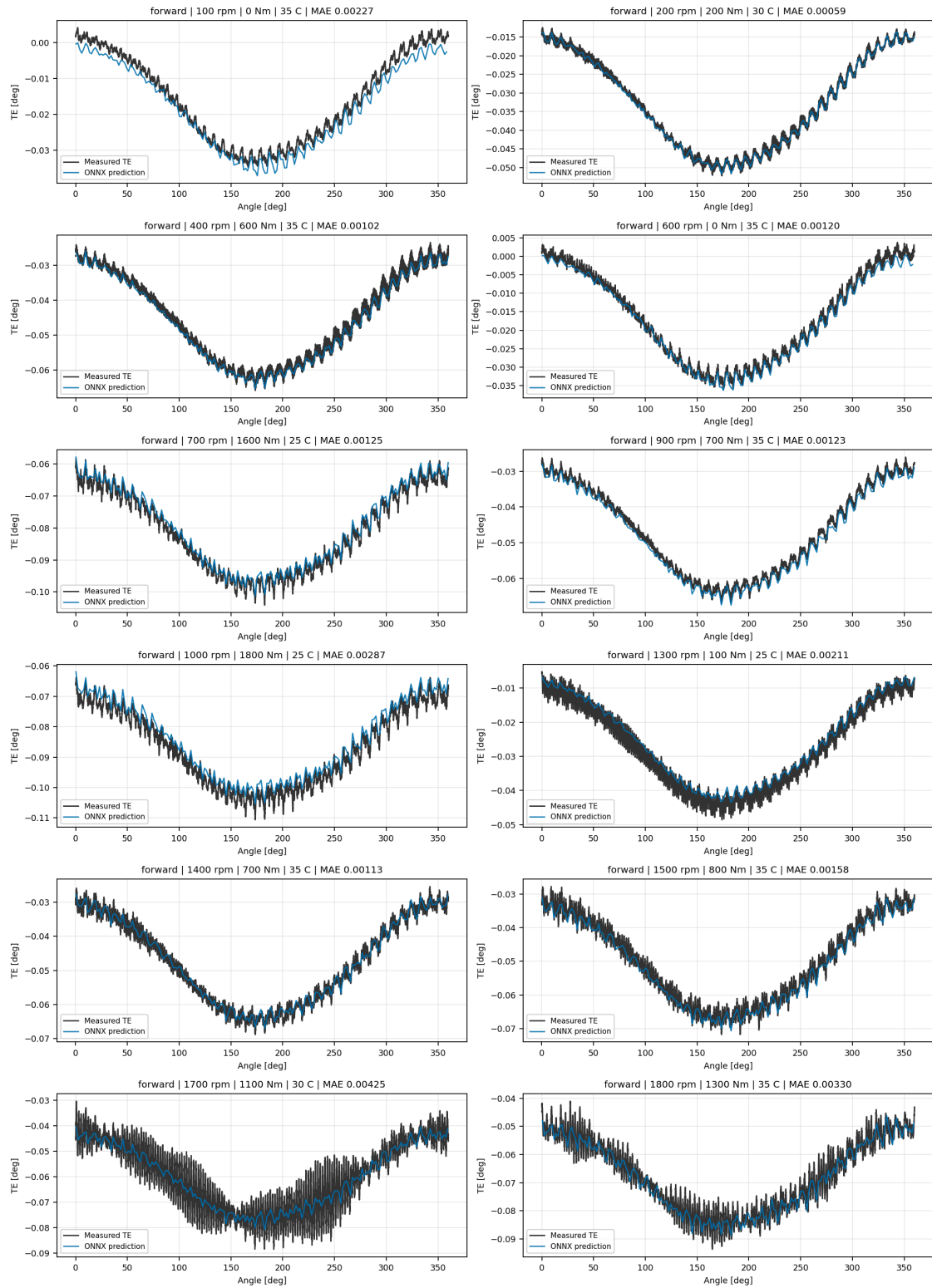
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001914	0.002238	4.210	10.391
backward	94	0.001671	0.001994	3.282	7.047
global	194	0.001561	0.001889	3.335	6.580

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000100	0.001100	0.001409	0.003072
backward	0.000352	0.000948	0.001263	0.003177
global	0.000003	0.000872	0.001230	0.002273

Forward 12-Curve Page

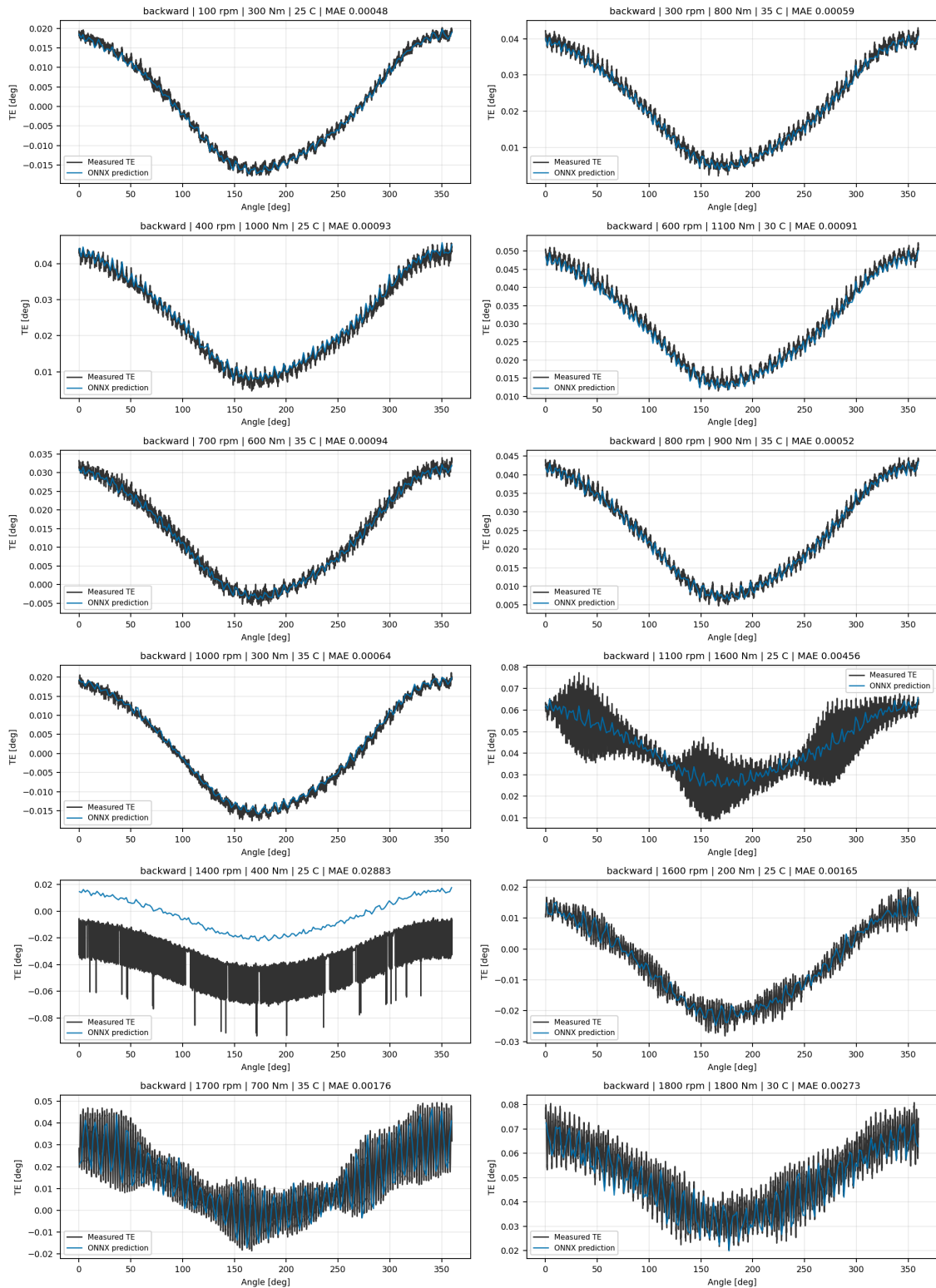
Polished Dataset + Setpoints | periodic_lstm_sequence | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

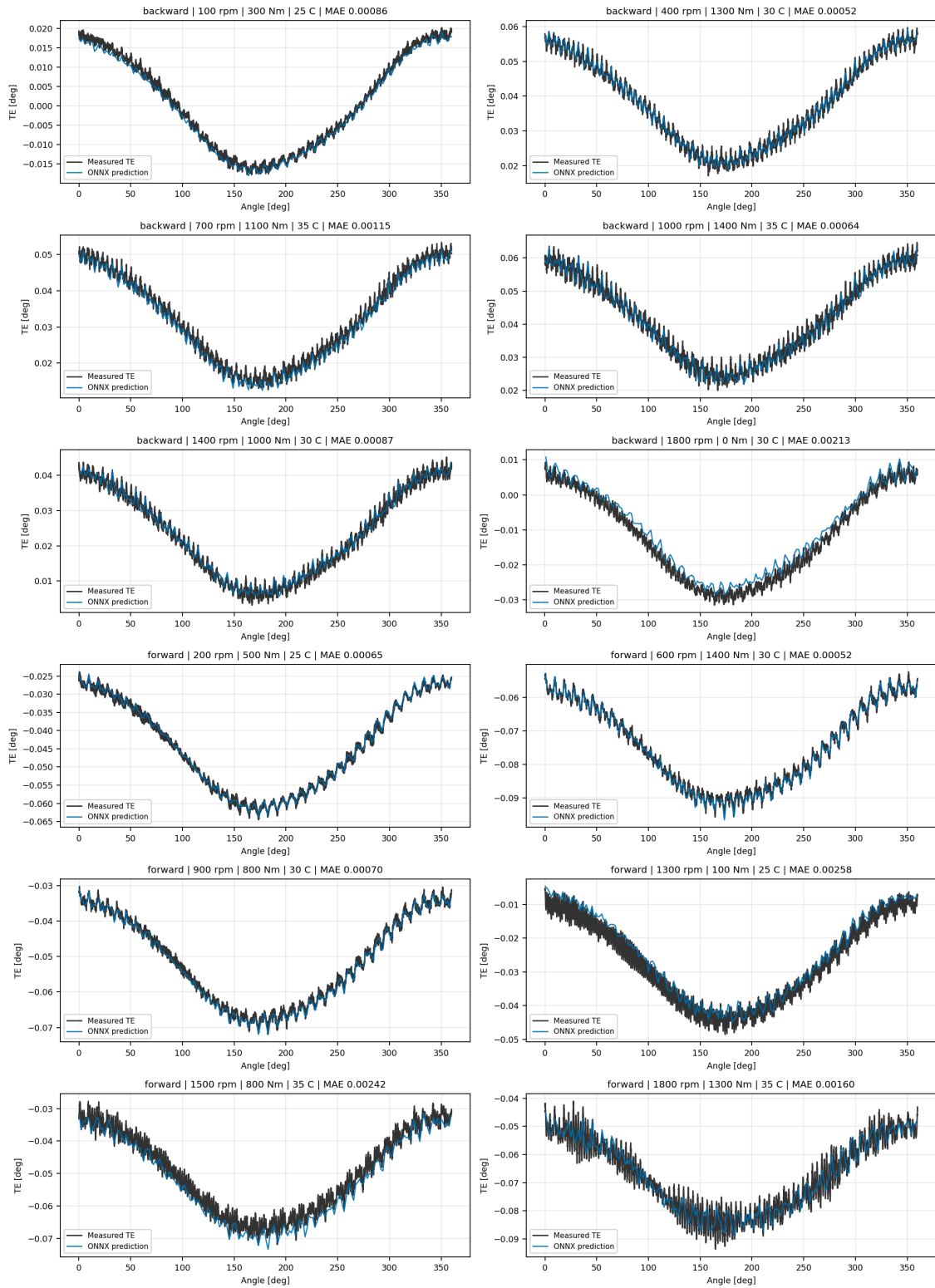
Polished Dataset + Setpoints | periodic_lstm_sequence | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | periodic_lstm_sequence | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: periodic_lstm_sequence ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_periodic_lstm_sequence_fw_polished_actual_values	2026-07-09-05-06-07__te_periodic_lstm_sequence_fw_polished_actual_values	polished_point_v1
backward	te_periodic_lstm_sequence_bw_polished_actual_values	2026-07-09-05-16-31__te_periodic_lstm_sequence_bw_polished_actual_values	polished_point_v1
global	te_periodic_lstm_sequence_global_polished_actual_values	2026-07-09-04-47-33__te_periodic_lstm_sequence_global_polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/periodic_lstm_sequence/forward/2026-07-09-05-06-07__te_periodic_lstm_sequence_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/periodic_lstm_sequence/forward/2026-07-09-05-06-07__te_periodic_lstm_sequence_fw_polished_actual_values/python/periodic_lstm_sequence-epoch=017-val_mae=0.00196587.ckpt
backward	models/polished_dataset/actual_values/exported/periodic_lstm_sequence/backward/2026-07-09-05-16-31__te_periodic_lstm_sequence_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/periodic_lstm_sequence/backward/2026-07-09-05-16-31__te_periodic_lstm_sequence_bw_polished_actual_values/python/periodic_lstm_sequence-epoch=053-val_mae=0.00197877.ckpt
global	models/polished_dataset/actual_values/exported/periodic_lstm_sequence/global/2026-07-09-04-47-33__te_periodic_lstm_sequence_global_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/periodic_lstm_sequence/global/2026-07-09-04-47-33__te_periodic_lstm_sequence_global_polished_actual_values/python/periodic_lstm_sequence-epoch=103-val_mae=0.00191666.ckpt

Aggregate Metrics

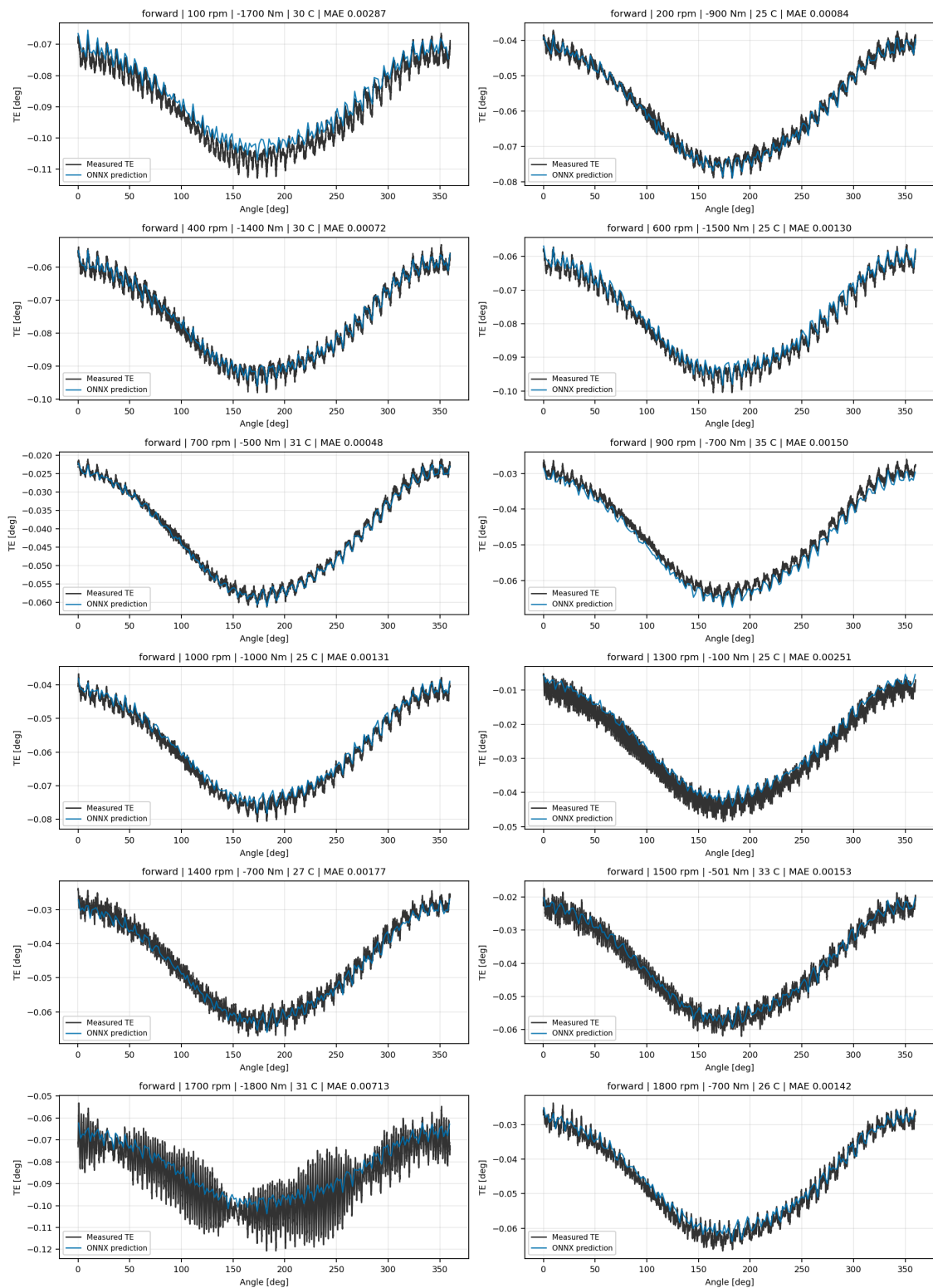
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001919	0.002246	4.216	10.613
backward	94	0.002539	0.002974	4.719	11.493
global	194	0.002121	0.002491	4.281	11.054

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000151	0.001013	0.001444	0.003569
backward	0.000517	0.000999	0.002114	0.005407
global	-0.000053	0.000961	0.001701	0.003590

Forward 12-Curve Page

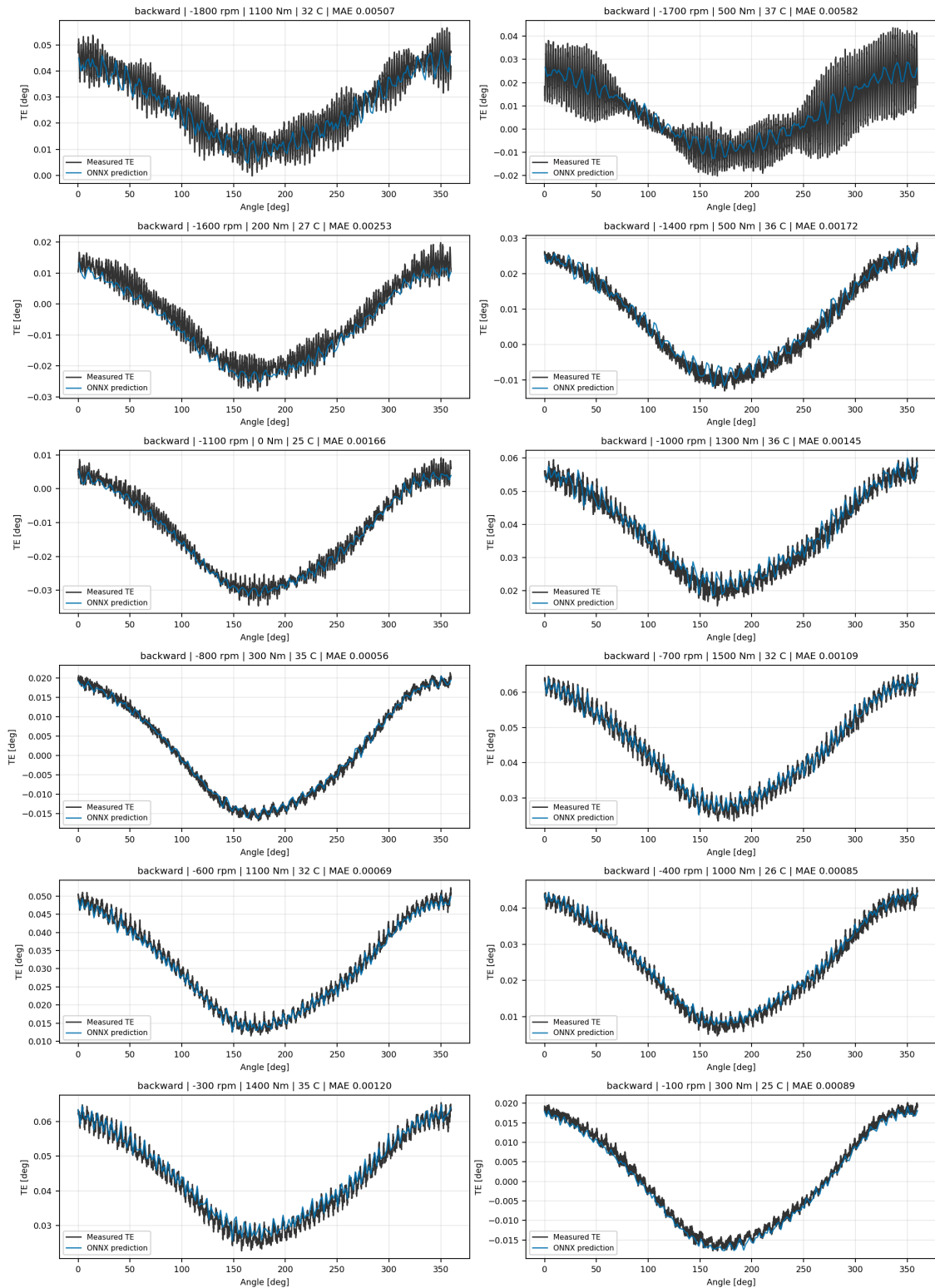
Polished Dataset + Actual Values | periodic_lstm_sequence | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

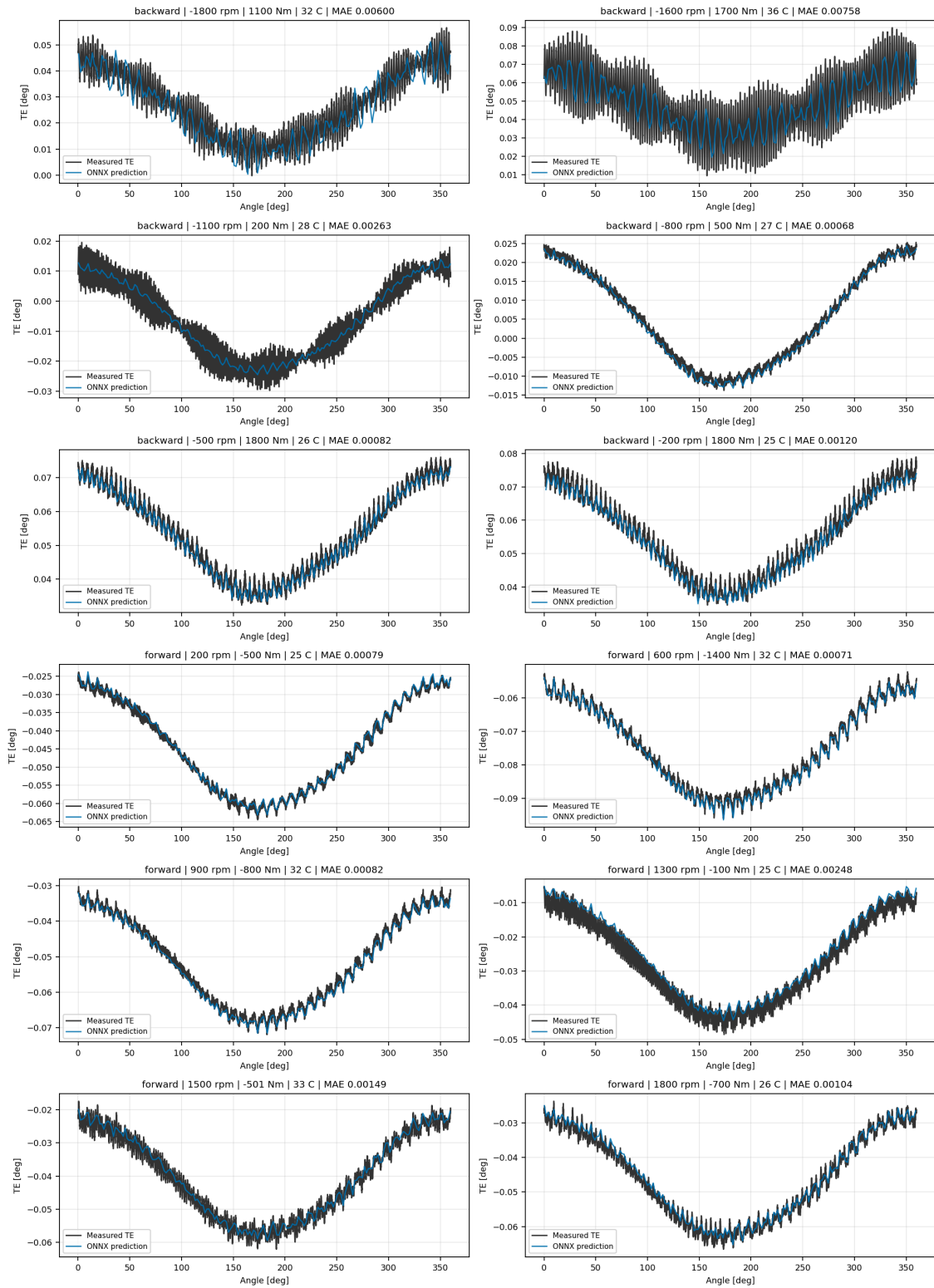
Polished Dataset + Actual Values | periodic_lstm_sequence | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | periodic_lstm_sequence | global



polished_dataset_actual_values global 12-curve collage